LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation on Current carrying capacity and effect of heat
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation on Current carrying capacity and effect of heat.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of wire gauge charts and calculation of heat generation based on current load; Instructor supervision required for safety calculations.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of current carrying capacity?
(b) How does electrical resistance relate to heat generation?
(c) What are the primary causes of aircraft wiring degradation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Current Carrying Capacity • Factors influencing wire ampacity • Relationship between current and heat • Insulation melting points and limits • Consequences of exceeding capacity	L & Ds	PPT, IB	15 min
(b)	• Joule's Law of Heating • Calculating heat dissipation in wires • Impact of ambient temperature • Bundle derating factors • Safety margins in aircraft wiring	L & Ds	PPT, PDF	20 min
(c)	• Common causes of wiring degradation • Thermal aging of insulation • Short circuit scenarios and heat • Preventive maintenance checks • Documentation of wire replacements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how excessive current leads to insulation failure.
(b) Describe three factors that reduce a wire's current carrying capacity.
(c) Why is derating necessary when multiple wires are bundled together?

12. Home Assignment and Follow Up	Prepare a brief note on the procedure for selecting the correct wire gauge for a repair, referencing the next lesson plan on wire substitution.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Wire Substitution for Repairs and Maintenance.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min